

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 4

SUBTRACTION OF TWO 8-BIT NUMBERS WITH BORROW

AIM: To write and execute an 8085 assembly language program to subtract two 8-bit numbers and store the sum and carry in memory.

ALGORITHM

1. Load the first number from 2500H.
2. Move it to register B.
3. Load the second number from 2501H.
4. Add register B with the accumulator.
5. Store the sum in 2502H.
6. Store the carry in 2503H.
7. Stop the program.

CODE:

```
LDA 2500H
MOV B,A
LDA 2501H
ADD B
STA 2502H
MVI A,00H
JNC NEXT
MVI A,01H
NEXT: STA 2503H
HLT
```

OUTPUT:

Registers

A	00
BC	00 00
DE	00 0A
HL	00 32
PSW	00 00
PC	42 16
SP	FF FF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2501

-

+

03

Update Memory

Load me at

1

LDA

2500H

2

MOV

B,A

3

LDA

2501H

4

ADD

B

5

STA

2502H

6

MVI

A,00H

7

JNC

NEXT

8

MVI

A,01H

9

NEXT:

STA

2503H

10

HLT

Data

Stack

Keypad

Memory

I/O Ports

Start

2500

OK

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	3
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG IN

23:48

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